

HDOC Day-23 - Completed

Topic: Square/Rectangular Patterns using nested loops, functions and switch statement in C

Aim: To display a menu and print the selected pattern using nested loops. Each pattern is implemented in a separate function with no return value.

Algorithm

1. Start the program.
2. Display a menu containing four pattern choices.
3. Read the user's choice.
4. Use a switch statement to call the corresponding function.
5. In the selected function, read the required number of rows/columns.
6. Use nested loops: the outer loop controls rows and the inner loop controls columns.
7. Print the required symbol/number for each position.
8. If the choice is invalid, display an error message.
9. Stop the program.

Test-Case Table

Test	Choice / Input	Expected Result
1	1, n = 4	4 x 4 solid square of *
2	2, m = 3, n = 5	3 x 5 solid rectangle of *
3	3, m = 4, n = 4	Rows: 1111, 2222, 3333, 4444
4	4, m = 4, n = 4	Each row: 1234
5	Choice = 7	Invalid choice message

C Program

```
#include <stdio.h>

void solidSquare(void);
void solidRectangle(void);
void rowNumberPattern(void);
void columnNumberPattern(void);

int main(void)
{
    int choice;

    printf("PATTERN MENU\n");
    printf("1. Solid square pattern\n");
    printf("2. Solid rectangular pattern\n");
    printf("3. Row-number pattern\n");
    printf("4. Column-number pattern\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            solidSquare();
            break;
        case 2:
            solidRectangle();
            break;
        case 3:
            rowNumberPattern();
            break;
        case 4:
            columnNumberPattern();
            break;
        default:
            printf("Invalid choice\n");
    }
}
```

```

        printf("Invalid choice!\n");
    }

    return 0;
}

void solidSquare(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n; j++)
            printf("* ");
        printf("\n");
    }
}

void solidRectangle(void)
{
    int m, n, i, j;
    printf("Enter rows and columns: ");
    scanf("%d %d", &m, &n);

    for (i = 1; i <= m; i++)
    {
        for (j = 1; j <= n; j++)
            printf("* ");
        printf("\n");
    }
}

void rowNumberPattern(void)
{
    int m, n, i, j;
    printf("Enter rows and columns: ");
    scanf("%d %d", &m, &n);

    for (i = 1; i <= m; i++)
    {
        for (j = 1; j <= n; j++)
            printf("%d ", i);
        printf("\n");
    }
}

void columnNumberPattern(void)
{
    int m, n, i, j;
    printf("Enter rows and columns: ");
    scanf("%d %d", &m, &n);

    for (i = 1; i <= m; i++)
    {
        for (j = 1; j <= n; j++)
            printf("%d ", j);
        printf("\n");
    }
}

```

Sample Outputs

1. Solid Square Pattern

```
Enter your choice: 1
Enter number of rows: 4
* * * *
* * * *
* * * *
* * * *
```

2. Solid Rectangular Pattern

```
Enter your choice: 2
Enter rows and columns: 3 5
* * * * *
* * * * *
* * * * *
```

3. Row-Number Pattern

```
Enter your choice: 3
Enter rows and columns: 4 4
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

4. Column-Number Pattern

```
Enter your choice: 4
Enter rows and columns: 4 4
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

Result

The C program was written using a switch statement, separate void functions and nested loops. The required square and rectangular patterns were generated successfully for the prepared test cases.